
A Better Dispersion

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Abstract

Since the rise in popularity of golf shot tracking hardware, golfers have been using their dispersion ellipse as a simple means to track performance statistics such as distance and shot consistency. Currently software fits a single ellipse to shot data via a single Gaussian model, but it is not always the case that a golfer's shot distribution can be accurately described by a single multivariate Gaussian. Our project explores using a Gaussian mixture model as an alternative to fit shot dispersion data. We also performed several experiments on the mixture model, evaluating how well it can be applied in a practice setting through techniques such as sampling, Monte Carlo simulation, and Markov Decision Processes. We found that the mixture model provides a better fit than a single Gaussian model, and we provide several examples of improved inference with the model.

1 Introduction

The goal of golf is to finish an 18 hole course in as few shots as possible. At any given time, picking the best shot is not as simple as choosing the one that minimizes the distance to the hole. Golfers must deal with hazards, course boundaries, and other obstacles on their way to the hole. Typically golfers use their dispersion pattern, represented by a single Gaussian model on shot data, to balance the risk and reward of a given shot, but this method is often purely visual and relies heavily on intuition. Though it can produce a numerical output, this is often not shown to golfers or well understood. Furthermore, it does not quantify the probability of adverse outcomes, and is sensitive to outliers and bias.

Our contribution is to provide golfers with a more accurate model of their shot data, that is easily accessible and personal to the average golfer, allowing them to simulate rounds and holes before they play them, leading to more informed decisions on the course, more targeted practice, better play, and an ultimate greater enjoyment of the game.

2 Related Work

A key advancement in measuring golf performance was brought about by the concept of strokes gained (Broadie 2012). Strokes gained is an attempt to measure the effect that a particular shot had on the resulting score of a hole. For example, a perfect shot down the center of the hole could contribute a stroke gained of 1, (1 for the actual stroke taken), where an imperfect shot could have a stroke gained of 1.3 or 1.7, implying that it will likely lead to a higher score on the hole than the perfect shot.

One drawback to this approach is that it does not predict what your expected score on a hole is. Drappi and Keh (2018) attempted to address this by providing a neural network with the task of using shot position and other data to predict the expected score on a hole. However, a further drawback of both methods is that the data used is that of professionals. Both methods used shots that professionals

hit in tournament play to get their labels, which were the expected strokes and the predicted score. For everyday golfers, this comparison is unrealistic.

Golf applications such as 18Birdies (2025) try to remedy this by comparing golfers of similar skill levels with one another, but as the skill level decreases, the variability of the shot increases and the predictions are not personalized to an individual golfer.

3 Methodology

3.1 Data / Data Collection

We want to make sure our data is easily accessible to golfers. As such, we choose to only use a golfer's carry distance ("Y") and lateral deviation ("X") as features. We used a combination of two datasets; the first comes from CaddieSet Jung et al. (2025), a dataset originally used to track the effect of joint positions on the shot result, but that also contains the two fields we need.

We also collected simulator data from Ryan (author) and a second UBC golfer. On the same simulator, Ryan also hit several shots on a par 3, recording the landing location to get empirical metrics to compare sampled results from our model. It took roughly 45 minutes to collect 60 shots from one of our UBC golfers.

3.2 Current Approach

The current approach for shot modeling is to use a single Gaussian model fit to the (X,Y) shot dataset. This model is often conveyed visually with an ellipse overlaid on a shot scatterplot, as shown in Figure 1. A single Gaussian model can be sensitive to outliers, and also cannot capture regions of higher density, as shown in figure Figure 1 where we can clearly see a dense region of shots along the target line ($x=0$), with a less dense region left of the target line.

3.3 Mixture Modeling Approach

Our approach fits a Gaussian mixture model to shot data, allowing for different mixtures to account for different miss areas or density regions within the shot data. We fit the model under two assumptions:

1. Any cluster of shot locations for a golfer is roughly normal. This assumption is probably met as off target misses are likely to vary by some unknown fixed amount based off of the mean miss location.

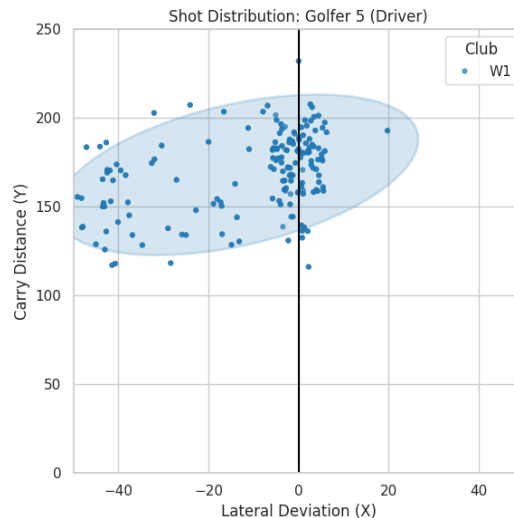


Figure 1: Golfer Dispersion Pattern 2 std Ellipse

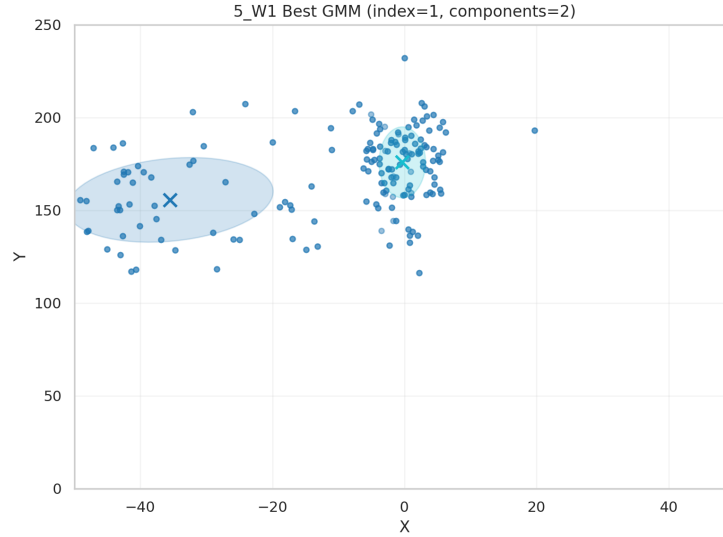


Figure 2: Sample Model Selected For Golfer (2 Components)

2. All golfer's shots are independent and identically distributed (i.i.d.). This assumption is less likely to be true. For example, a miss left could cause a golfer to aim right on the next shot to account for the miss. Since we are primarily interested in using the model for real world inference, it is fine if this assumption is not fully met as the shot corrections mentioned previously would also be made in a practical setting.

To select the number of components in the mixture, we used a 90/10 training validation split. For each golfer and each club, we fit a mixture with 1-10 components and compared models using both Energy Distance on the validation set and the Bayesian Information Criterion (BIC), as a primary goal of the model is accurate inference. Specifically, we first selected the top 5 models with the lowest Energy Distance, and then chose the model with the lowest BIC among those. Figure 2 shows the model selected for this golfer, which in this case has 2 components

4 Experimentation

4.1 Sampling

Much of our later inference and applications depend on the sample quality of our mixture. Figure 3 shows the shot distribution of a golfer in their selected model, with contours on the logarithmic scale. Visual inspection indicates a higher probability density in the central target cluster, with a lower—but still significant—probability in the left, more diffuse cluster.

We also compared samples of individual shots from both the current single-component Gaussian model and our mixture model to the actual data. Figure 4 shows these results. The samples from the mixture model match the structure of the observed data more closely than those from the single Gaussian model. The mixture was able to capture both clusters of shots and maintain relative densities of the clusters, whereas the single Gaussian produced a single uniformly dense cluster of points across the range of the data.

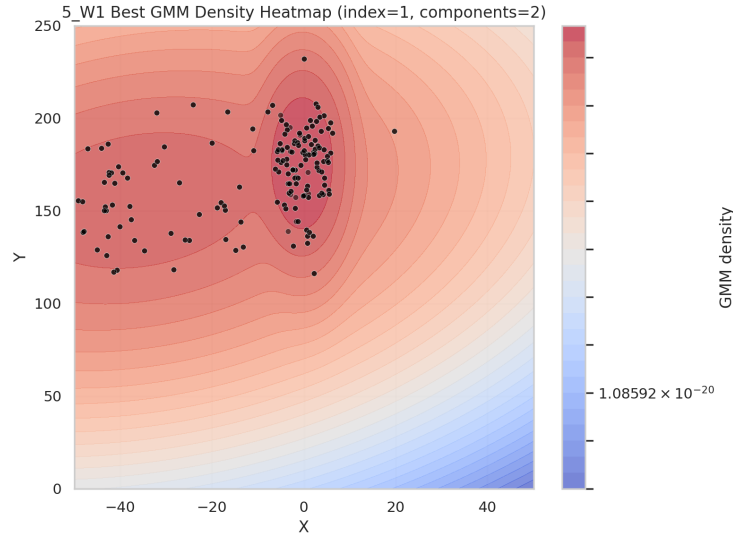


Figure 3: Distribution Heat Map (Density on log scale)

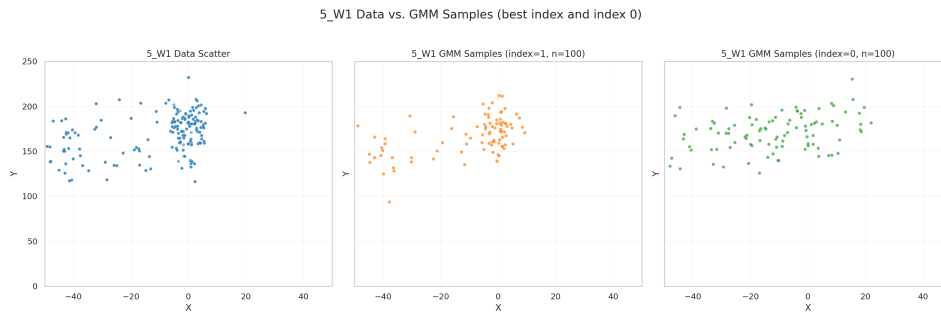


Figure 4: Original Data (Left) Samples from GMM (Middle) Samples From Current Method (Right)

86 4.2 Monte Carlo

87 Monte Carlo simulation can be used to assess the probability of landing in hazards when hitting
 88 certain shots. Figure 5 shows a simulated golf hole, with shots sampled from Ryan's fitted mixture
 89 model. (Hole 9 Hawk Priddis Greens Golf and Country Club). The sampled results closely match the
 90 true results of Ryan playing this hole on a golf simulator.

91 Table 1 shows the probabilities of landing on the green, in the water, or in the bunker over several
 92 shots. The simulated probabilities are reasonably consistent with the empirical results, suggesting
 93 that the model captures the underlying shot distribution with acceptable accuracy.

94 These simulations can be used to potentially inform golfers about different target lines. For example,
 95 if a golfer was willing to reduce the probability of landing on the green in exchange for a lower risk
 96 of hazards, they could adjust their target, rerun the simulation, and have numerical evidence to guide
 97 their decision.

Table 1: Simulated vs Actual Landing Probabilities

Location	Simulated Probability	Actual Probability
Green	0.400	0.500
Water	0.120	0.090
Bunker	0.140	0.170

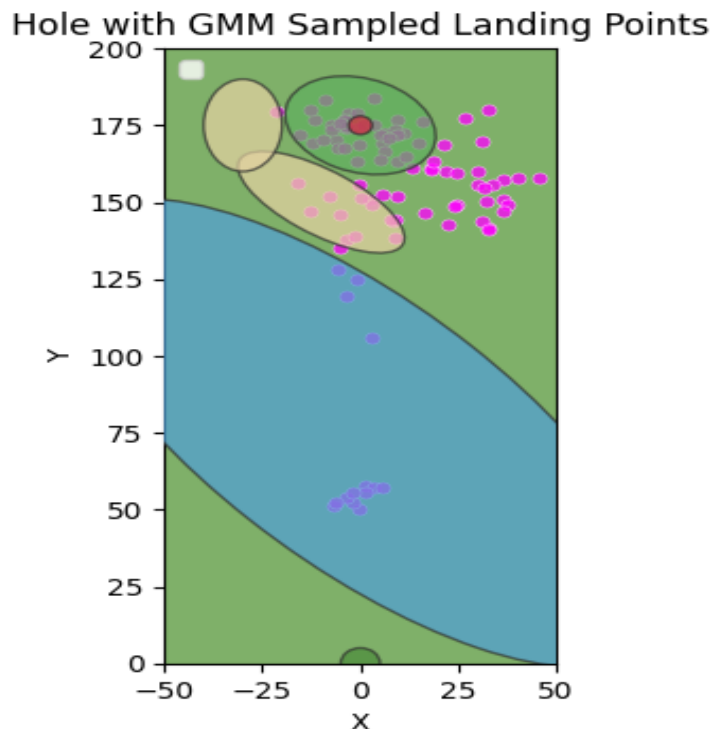


Figure 5: Golfer Shot Samples On Simulated Hole

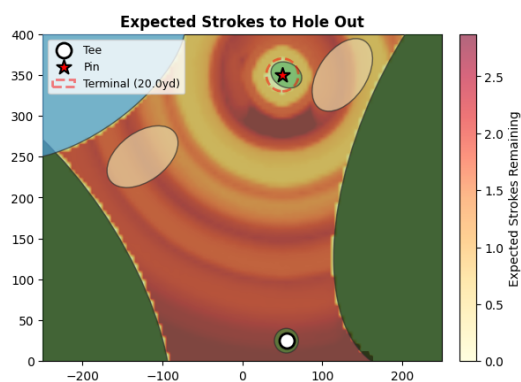


Figure 6: Heat Map of Expected Hole Score MDP

98 4.3 Markov Decision Process

99 The last of our experiments came in the form of using our Gaussian mixture models as a transition
 100 model in a Markov Decision Process with states represented by the (x,y) position of a golfer ball
 101 on a hole, and actions represented by (state, club, target) tuples. The goal of the MDP would be to
 102 recommend a club and target for the golfer to hit and aim at given their current position that would
 103 minimize the expected score on a hole. We utilized the standard value iteration technique and a
 104 greedy algorithm to obtain our policy.

105 The heatmap in Figure 6 show the expected score in strokes on a simulated hole using a limited
 106 selection of clubs. The clubs provided were a real driver, a real 7 iron, and a simulated short wedge,
 107 allowing for MDP convergence. We can see in the image given the available clubs, there are rings
 108 around the target that give lower expected scores, since a hole distance that is in between the distances
 109 of clubs could lead to a higher score.

110 In the simulation, for every shot taken a reward of -1 was given, representing a stroke. Shots into
111 trees or landing in water were simplified with a 2 stroke penalty and replacement of the ball, similar
112 to what would happen in real life.

113 The MDP can also be combined with Monte Carlo methods, providing an distribution of expected
114 scores on a hole.

115 5 Discussion / Future Work

116 Our work shows that the standard dispersion ellipse can be greatly improved through the use of a
117 mixture model. We have shown that even with limited data, this model is a more accurate description
118 of a golfer shot distribution and that it can generate more accurate samples. We have also shown
119 applications of this model, with Monte Carlo simulations, and Markov Decision Processes. Our work
120 lays a foundation for a new way to look at and utilize shot data. In the future the MDP should be
121 explored further. We believe that the Markov decision process can be optimized for more complex
122 holes, more realistic actions, and include different state descriptors such as ball lie.

123 Acknowledgments

124 Thank you to Griffin McDonald, the fellow UBC golfer who helped collect data.

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136 A Supplementary material

137 A.1 Code

138 If interested our code repository can be found here

139 A.2 Golfer All Components

140 Figure 7 shows mixtures with 1-7 components for the golfer discussed.

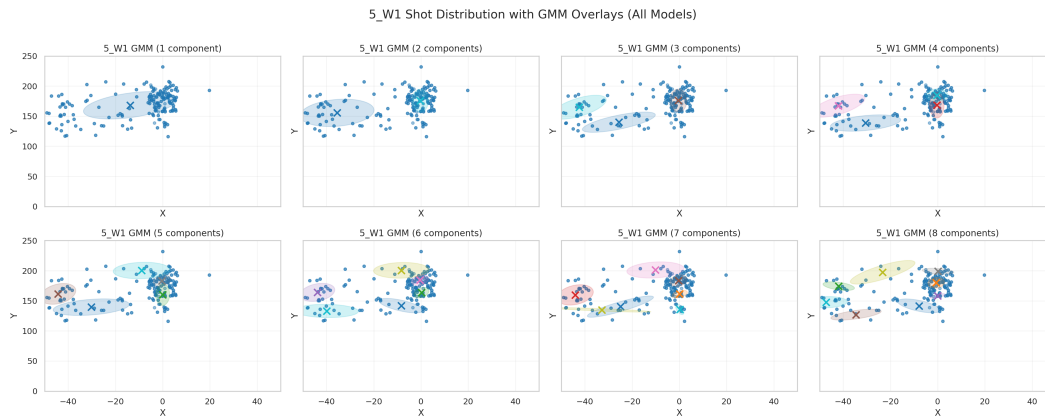


Figure 7: 1-7 Component mixtures for a golfer